

- (iii) After 6 months of exercising, the mean and standard deviation of these 80 persons are 50kg and 12.2 kg respectively. Give 2 comments on the effect of exercise.

1

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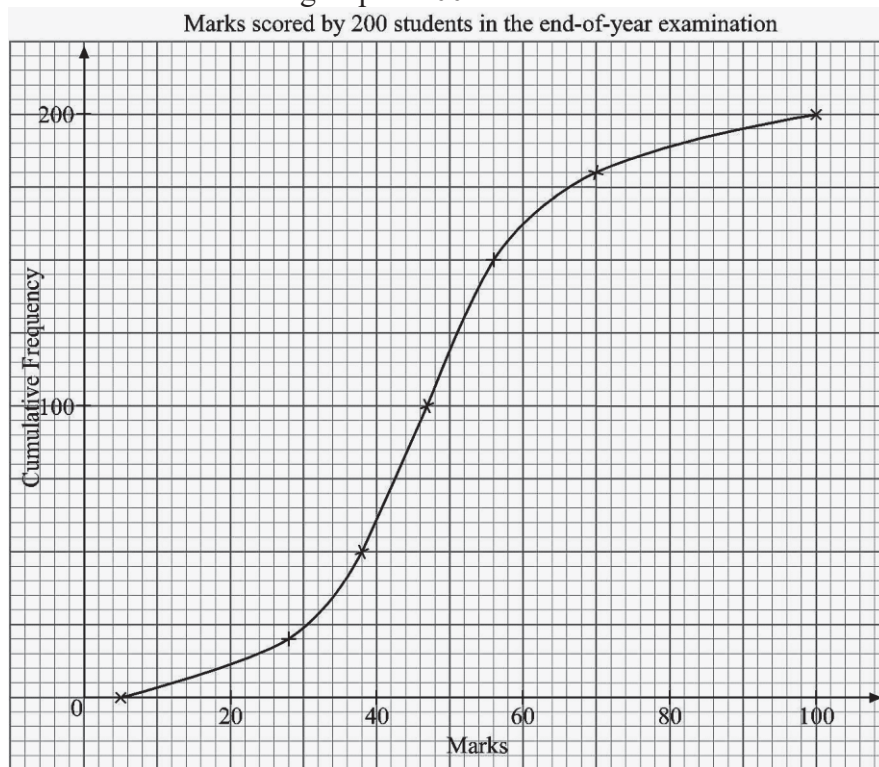
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2

.....

..... [2]

- 10 (c) The cumulative frequency graph below shows the end-of-year Mathematics examination marks of a group of 200 students.



From the graph, find the

- (i) number of students who scored 28 marks or less,

Answer [1]

- (ii) number of students who scored more than 76 marks,

Answer [1]

- (iii) the upper quartile,

Answer [1]

- (iv) minimum mark obtained by the top 2.5% of the cohort,

Answer [1]

- (v) probability that one student scored 28 marks or less and the other student scored more than 76 marks when two students are chosen at random.

Answer[2]

- 11 The variables x and y are connected by the equation

$$y = \frac{5x}{4} + \frac{1}{x^2}.$$

The table below shows some values of x and the corresponding values of y , correct to 1 decimal place.

x	0.5	1	1.5	2	3	4	5
y	4.6	2.3	2.3	2.8	3.9	5.1	p

- (a) Calculate the value of p .

Answer $p =$ [1]

- (b) Using a scale of 2 cm to represent 1 unit on each axis, draw a horizontal x -axis for $0 \leq x \leq 5$ and a vertical y -axis for $0 \leq y \leq 8$. On your axes, plot the points given in the table and join them with a smooth curve. [3]

- (c) From the graph, find the values of x for which $\frac{5x}{4} + \frac{1}{x^2} - 3 = 0$.

Answer $x =$ or..... [2]

- (d) By drawing a tangent, find the gradient of the curve at the point $(1, 2.3)$.

Answer [2]

- (e) (i) On the same axes, draw the graph of $y = \frac{1}{2}x + 2$. [2]

- (ii) Write down the x -coordinates of the points where the two graphs intersect.

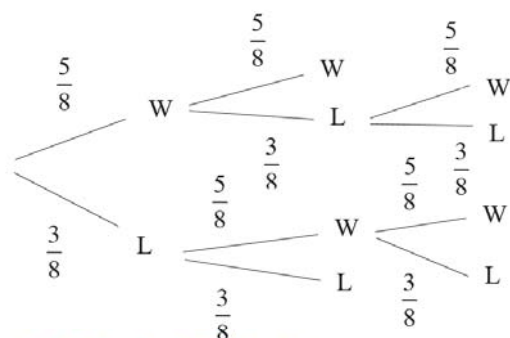
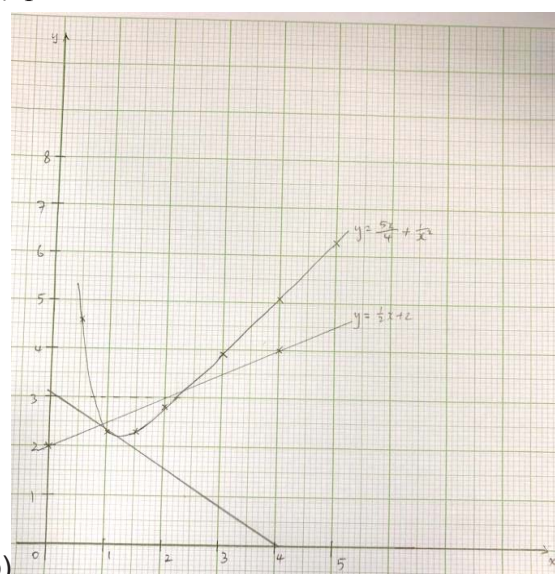
Answer $x =$ or..... [1]

- (iii) Find the equation in the form $3x^3 + ax^2 + bx + c = 0$, which is satisfied by the values of x found in part (e)(ii).

Answer [1]

END OF PAPER

Answer key for paper 2

1	(a)(i) $(2p+3q)(x-1)$ (ii) $x=1$ (b) $\frac{9q^4}{p^2}$	2	(a) 1.29 radians (b) 4.09cm^2
3	(b) A(0, 9). D(0, 3) (c) E(5, 15) (d) 45 unit^2	4	(a) (i) $\$ \frac{65}{x}$ (ii) $\$ \frac{65}{x-6}$ (c) $x=17.28$ or $x=-11.28$ (d) \$5.76 (to nearest cent)
5	(b)(i) 061.6° (ii) 4.41 km (c) 0.231 km.		
7	(a) (i) 74° (ii) 74° (iii) 41° (b) (i) 13 (Tangent from ext. point) (ii) 6 cm	6	(a) 0.0015, 0.0018, 0.225(egg) (b) $\mathbf{E} = \begin{pmatrix} 0.0015 \\ 0.0018 \\ 0.225 \end{pmatrix}$ (c) $\begin{pmatrix} 0.5625 \\ 2.49 \\ 0.684 \end{pmatrix}$ (c) The elements in DE represents the cost price (in dollars) needed for the making of one cookie, one cake and one pancake respectively (d) \$196.16
8	(a) (ii) $130\frac{2}{3}\text{mm}^2$ or $131\text{mm}^2(3\text{sf})$ (iii) 64.4mm^2 (b) $42.9\text{cm}(3\text{sf})$		
9	(a)(i) \$26,399.70 (ii) \$1847.36 (b) \$924 (Accept answers from \$939 to \$1064.40 as some students may assume up to 23 working days and 4 weekends or 21 working days and 5 weekends) (c) Acceptable responses such as 1. Renting is better as there is no need to pay down payment and monthly instalment. 2. Although it is more expensive to buy a car, it is very convenient. 3. Any other reasonable justification.		
10 (a)	(a)(i)  (ii) $\frac{175}{256}$ (b) (i) 53.875 kg or $\frac{431}{8}\text{kg}$ or $53\frac{7}{8}\text{kg}$ (ii) 9.87 kg (iii) (1) The mean weight is lower. Jogging has helped to reduce weight. (2) Standard deviation is higher implies that weight loss after exercising is more spread out. (accept "not consistent") (c) (i) 20 students (ii) 15 students (iii) 56 marks (iv) 90 marks (v) $\frac{3}{199}$	11	(a) $p \approx 6.3$  (b) (c) From the graph, when $y=3$, $x=0.7$ or $x=2.2$ (Accept ± 0.20) (d) From the tangent at (1, 2.3), the gradient is -0.75 (± 0.2) (e) (i) Draw graph of $y = \frac{1}{2}x + 2$ (ii) The x coordinates are 0.9 and 2.3 (iii) $3x^3 - 8x^2 + 4 = 0$